Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY **9700/33**

Paper 3 Advanced Practical Skills 1 **February/March 2020**

CONFIDENTIAL INSTRUCTIONS

7135002886

This document gives details of how to prepare for and administer the practical exam.

The information in this document and the identity of any materials supplied by Cambridge International are confidential and must NOT reach candidates either directly or indirectly.

The supervisor must complete the report at the end of this document and return it with the scripts.

INSTRUCTIONS

- If you have any queries regarding these confidential instructions, contact Cambridge International stating the centre number, the syllabus and component number and the nature of the query.
  email info@cambridgeinternational.org
  phone +44 1223 553554

This document has **8** pages. Blank pages are indicated.

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General information about practical exams

Centres must follow the guidance on science practical exams given in the *Cambridge Handbook*.

Safety

Supervisors must follow national and local regulations relating to safety and first aid.

Only those procedures described in the question paper should be attempted.

Supervisors must inform candidates that materials and apparatus used in the exam should be treated with caution. Suitable eye protection should be used where necessary.

The following hazard codes are used in these confidential instructions, where relevant:

C	corrosive	**MH**	moderate hazard
HH	health hazard	**T**	acutely toxic
F	flammable	**O**	oxidising
N	hazardous to the aquatic environment		

Hazard data sheets relating to substances used in this exam should be available from your chemical supplier.

Before the exam

- The packets containing the question papers must **not** be opened before the exam.
- It is assumed that standard school laboratory facilities, as indicated in the *Guide to Planning Practical Science*, will be available.
- Spare materials and apparatus for the tasks set must be available for candidates, if required.

During the exam

- It must be made clear to candidates at the start of the exam that they may request spare materials and apparatus for the tasks set.
- Where specified, the supervisor **must** perform the experiments and record the results as instructed. This must be done **out of sight** of the candidates, using the same materials and apparatus as the candidates.
- Any assistance provided to candidates must be recorded in the supervisor's report.
- If any materials or apparatus need to be replaced, for example, in the event of breakage or loss, this must be recorded in the supervisor's report.

After the exam

- The supervisor must complete a report for each practical session held and each laboratory used.
- Each packet of scripts returned to Cambridge International must contain the following items:
    - the scripts of the candidates specified on the bar code label provided
    - the supervisor's results relevant to these candidates
    - the supervisor's reports relevant to these candidates
    - seating plans for each practical session, referring to each candidate by candidate number
    - the attendance register.

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Specific information for this practical exam

During the exam, the supervisor or other competent biologist (**not** the invigilator) should obtain the results specified on the supervisor's report by following the relevant steps in the question paper. The results should be recorded in the space provided on the supervisor's report.

Organisation of the exam

- Half the candidates should start on Question 1 and the other candidates should start on Question 2.
- For Question 2, each candidate must have uninterrupted use of a microscope for at least 55 minutes.

Materials to be supplied by Cambridge International

- None

Materials and apparatus for Question 1

Each candidate will need:

materials and apparatus for each candidate	quantity	✓
3% yeast suspension in a beaker suitable for heating, labelled **Y**, provided at room temperature (see **Preparation of materials**)	at least 100 cm^3	
3% hydrogen peroxide solution in a beaker, labelled **H**, provided at room temperature (see **Preparation of materials**)	at least 20 cm^3	
10 cm^3 syringe	1	
Test-tube, large, capacity 40 cm^3 to 55 cm^3	1	
Test-tube rack to hold one large test-tube	1	
Bunsen burner, tripod, gauze, bench mat and lighter or matches	1	
Glass rod	1	
Container with approximately 200 cm^3 of tap water, labelled **For washing**	1	
Container, capacity approximately 200 cm^3, labelled **For waste**	1	
Paper towels	8	
Glass marker pen, permanent	1	
Stop-clock or timer showing seconds	1	
Thermometer, –10 °C to +110 °C	1	
Suitable eye protection	1	

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Preparation of materials

- **Y**, 3% yeast suspension

  This is prepared on the day of the exam. Put 3 g of yeast into a beaker and make up to 100 cm^3 with distilled water. Mix well.

  If preparing large batches, mix well before distributing into beakers for individual candidates.

- **H**, 3% hydrogen peroxide solution

  This is prepared on the day of the exam. Put 50 cm^3 of 6% (20 vol) hydrogen peroxide solution **[MH]** into a beaker and add 50 cm^3 of distilled water. Mix well.

Materials and apparatus for Question 2

Each candidate will need:

materials and apparatus for each candidate	quantity	✓
Microscope with: ● an eyepiece lens, ×10 magnification ● a low-power objective lens, ×10 magnification ● a high-power objective lens, ×40 magnification	1 between 2	
1.5% potato starch in suspension in a beaker, labelled **S**, provided at room temperature (see **Preparation of materials**)	at least 15 cm^3	
Clean, dry microscope slide with coverslip	1	
Paper towels	8	
Pasteur pipette	1	
Glass rod	1	
Mounted needle	1	
Glass marker pen, permanent	1	
Ruler, marked in mm	1	

Preparation of materials

- Microscope

  Any lenses which are **not** ×10 or ×40 should be removed or replaced.

  For each candidate:

  - the microscope must be set up on low power
  - the slide must **not** be on the stage of the microscope.

- **S**, 1.5% potato starch suspension

  This is prepared by putting 1.5 g of potato starch into 100 cm^3 of distilled water and stirring well.

  Do not use soluble potato starch.

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Supervisor's report

Syllabus and component number	9 7 0 0 / 3 3
Centre number	

Centre name .....

Time of the practical session .....

Laboratory name/number .....

Give details of any difficulties experienced by the centre or by candidates (include the relevant candidate names and candidate numbers).

You must include:

- any difficulties experienced by the centre in the preparation of materials
- any difficulties experienced by candidates, e.g. due to faulty materials or apparatus
- any specific assistance given to candidates.

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Temperature of exam room ..... °C

Results for Question **1(a)(iv)**

Declaration

1 Each packet that I am returning to Cambridge International contains the following items:

- ☐ the scripts of the candidates specified on the bar code label provided
- ☐ the supervisor's results relevant to these candidates
- ☐ the supervisor's reports relevant to these candidates
- ☐ seating plans for each practical session, referring to each candidate by candidate number
- ☐ the attendance register.

2 Where the practical exam has taken place in more than one practical session, I have clearly labelled the supervisor's results, supervisor's reports and seating plans with the time and laboratory name/number for each practical session.

3 I have included details of difficulties relating to each practical session experienced by the centre or by candidates.

4 I have reported any other adverse circumstances affecting candidates, e.g. illness, bereavement or temporary injury, directly to Cambridge International on a *special consideration form*.

Signed ..... (supervisor)

Name (in block capitals) .....

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